

1 From Constitutional Purpose to Accelerated Verified Learning

AION now contains a connected schooling architecture rather than a collection of independent capability demonstrations. The architecture binds an immutable constitutional purpose to an evidence-backed capability registry, a governed teacher–learner–examiner cycle, a persistent curriculum supervisor, a depth-first technical academy, executable assessment, and compressed retained intelligence. Its complete control path is

$$\begin{aligned} \text{constitutional purpose} &\rightarrow \text{capability and weakness map} \rightarrow \text{strategic academy} \\ &\rightarrow \text{teacher-advised curriculum} \rightarrow \text{executable practice} \\ &\rightarrow \text{sealed assessment} \rightarrow \text{targeted remediation} \\ &\rightarrow \text{transfer and retention} \rightarrow \text{governed intelligence glyphs.} \end{aligned} \tag{1}$$

The principal retained procedures are

- `procedure_constitutional_north_star_mastery_registry_v1`;
- `procedure_governed_teacher_python_core_cycle_v1`;
- `procedure_autonomous_mastery_curriculum_supervisor_v1`;
- `procedure_guided_programming_systems_security_academy_v1`;
- `procedure_accelerated_algorithms_data_structures_apprenticeship_v1`; and
- `procedure_governed_intelligence_glyph_consolidation_v1`.

1.1 Constitutional North Star and Capability Authority

The North Star imposes a persistent hierarchy

$$\begin{aligned} \text{constitutional purpose} &\rightarrow \text{maturity stage} \rightarrow \text{strategic learning portfolio} \\ &\rightarrow \text{autonomous goals} \rightarrow \text{governed action.} \end{aligned} \tag{2}$$

The constitutional purpose is immutable within the runtime. It directs AION to develop and use increasingly general intelligence for beneficial, evidence-backed capability while respecting human authority, law, consent, safety, provenance, and legitimate uncertainty. Resources, access, money, influence, infrastructure, and self-preservation are explicitly prohibited from becoming terminal objectives.

The Mastery Registry reconstructs competence from verified artifacts rather than from self-description. It distinguishes six levels:

$$\mathcal{L} = \{\text{unassessed}, \text{learning}, \text{operational}_b, \text{proficient}_b, \text{expert}, \text{mastered}\}, \tag{3}$$

where the subscript b denotes a bounded claim. Expert and mastered status cannot be inferred merely from an internally promoted procedure.

Table 1: Evidence-backed subject status at the recorded North-Star snapshot.

Level	Subjects
Proficient, bounded	English; Python; Rust; scientific method; security engineering
Operational, bounded	Mathematics; JavaScript/TypeScript; SQL and databases; software engineering; data and statistics; business management
Learning	Physics; finance and accounting; UI/product design; creative invention
Unassessed	Spanish; biology and medicine; chemistry; social/commonsense intelligence; Go; Java; C/C++; web platform
Expert	None
Mastered	None

The registry answers operational questions, not only produces a score. For example, “Do you know Rust?” returns `yes_bounded`; “Can you do a Rust job?” returns `yes_with_limits`; and “Have you mastered Rust?”

returns `no` together with missing evidence. Each answer carries demonstrated subskills, procedure identifiers, outcome authorities, artifact hashes, limitations, and the conditions under which work may safely proceed.

At this snapshot AION tracked 23 subjects across nine groups. Eleven subjects were operational or better and five of those were non-computational. The apprentice retained five transfer families, four executor contracts, and five verified apprenticeship receipts. The General Apprentice gate remained closed because it required at least six transfer families, five executor contracts, and ten verified receipts. Unsupported expert and mastery claims were both zero; unsafe actions and objective mutations were also zero.

1.2 Governed Teacher–Learner–Examiner Separation

The first schooling cycle establishes a strict separation of roles:

- a replaceable language model proposes curricula, explanations, and practice;
- the apprentice studies, acts, and proposes solutions;
- independent execution, hidden tests, transfer, and retention grade competence; and
- HexCore and CAU authorize promotion.

The teacher cannot see the sealed answer key, grade its own teaching, or convert linguistic plausibility into knowledge.

The assessment contract is content-addressed and committed before study. Let s_i be the score for component i and w_i its declared weight. The overall score is

$$S = 0.15s_K + 0.30s_C + 0.20s_D + 0.15s_{TS} + 0.15s_X + 0.05s_R, \quad (4)$$

where K is knowledge, C construction, D debugging, TS testing and security, X source-disjoint transfer, and R retention. Promotion requires $S \geq 0.90$, all category minima, a complete security pass, fresh-process execution, unfamiliar transfer, restart reconstruction, and zero live source writes.

The local `gemma3:1b` model was consulted through Ollama as a real teacher. Its response was cached and recorded with provenance. The curriculum critic retained useful object-oriented, database, and systems-design themes but found that the proposal did not adequately specify typing contracts, error/resource management, transfer, or delayed retention. It was therefore retained only as a supplement; it could neither replace the core curriculum nor authorize a certificate.

The initial Python candidate illustrates why the separation matters. It passed public examples but treated booleans as valid domain integers because `bool` is a Python subclass of `int`. The sealed assessment exposed the error. AION attributed the failure to one input-semantics subskill, remediated only that subskill, and passed a fresh sealed variant. It then transferred the input-and-state invariant into an unrelated account-processing application and rejected six malicious programs before execution.

Table 2: Governed Python Core assessment-cycle results.

Measure	Result
Curriculum modules proposed	12
Exam committed before execution	Passed
Initial plausible candidate rejected	Passed
Targeted remediation	Passed
Fresh sealed assessment	Passed
Source-disjoint transfer	Passed
Malicious programs rejected	6/6
Restart reconstruction	Passed
Unsafe executions/live writes	0/0
Focused tests	10/10

The certificate `certificate_python_core_governed_v1` is deliberately `operational_bounded`. Its 100% score describes the bounded assessment cycle, not all of Python. No Python specialization, expert status, or language mastery is authorized, and an elapsed-time retention test remains due.

1.3 Persistent Autonomous Curriculum Supervision

The schooling mechanism is not limited to manually launched runs. A persistent service repeatedly executes

$$\begin{aligned} \text{North Star} &\rightarrow \text{select gap} \rightarrow \text{consult teacher} \\ &\rightarrow \text{commit curriculum contract} \rightarrow \text{match executor and authority} \\ &\rightarrow \text{learn, or request missing machinery.} \end{aligned} \tag{5}$$

The supervisor imported the Python certificate, scheduled a fresh delayed retest with solution replay forbidden, and autonomously selected social/commonsense intelligence, biology/medicine, and chemistry as its first three broad gaps. Three live Gemma consultations were completed with zero owner-supplied lesson steps.

These subjects were not falsely marked learned. Social competence requires independent multi-rater outcomes and retention of legitimate disagreement; biology and chemistry require authoritative evidence plus withheld observations or safe experimental outcomes. The supervisor therefore created three persistent executor-acquisition requests. This is a fail-closed distinction between knowing what should be learned and possessing an authority capable of proving it.

The original breadth-first selection was subsequently parked because it could produce fragmented subject hopping. The contracts remain retained for later use, but they no longer consume the primary learning schedule.

1.4 Guided Programming, Systems and Security Academy

The guided academy provides the accelerated middle layer between passive training-data ingestion and slow open-ended rediscovery. The owner supplies a coherent mastery graph and high-quality learning capsules. AION still selects practice, requests instruction, diagnoses failures, adjusts difficulty, constructs remediation, and learns from outcomes.

The resource allocation is

$$(\rho_P, \rho_R, \rho_X) = (0.70, 0.20, 0.10), \tag{6}$$

where P is primary academy progression, R retention/remediation, and X cross-domain transfer. The academy comprises the following prerequisite graph:

1. Python Core;
2. Algorithms and Data Structures;
3. Advanced Python;
4. Testing and Debugging;
5. Software Engineering;
6. Database Engineering;
7. Operating Systems;
8. Networking;
9. Rust Systems Engineering;
10. Secure Software Engineering;
11. Distributed Systems;
12. Architecture and Production Operations; and
13. an Integrated Engineering Capstone.

Python Core was imported as the first bounded certificate. Algorithms, Advanced Python, and Testing/Debugging then became eligible according to their prerequisites. Software Engineering remains locked until Algorithms, Advanced Python, and Testing/Debugging are all verified. The integrated capstone is the only path to a broad engineering-proficiency claim; completion of a single language module is insufficient.

The acceleration mechanism is

$$\begin{aligned} \text{curated capsules} &\rightarrow \text{executable practice} \rightarrow \text{verified trajectories} \\ &\rightarrow \text{semantic and skill glyphs} \rightarrow \text{neural consolidation.} \end{aligned} \tag{7}$$

Curated material reduces resource-discovery cost, but it does not bypass assessment, transfer, or retention.

1.5 Accelerated Algorithms and Data Structures Apprenticeship

The first academy module used six compact learning capsules: complexity growth, binary search, breadth-first search, weighted shortest paths, topological ordering, and dynamic programming. Each capsule carried a governing principle, a characteristic misconception or counterexample, and an executable practice contract.

The initial candidate passed public tasks but accepted a negative-weight edge inside Dijkstra’s algorithm. The sealed examiner rejected the solution because Dijkstra’s greedy finalization requires non-negative weights. AION localized the failure to the missing precondition rather than restarting the complete course. The remediated challenger passed a fresh sealed cohort and 120 generated property cases.

Topological ordering then transferred into an unfamiliar project-dependency problem. The method had to place every prerequisite before its dependent task, detect cycles, survive reconstruction, and pass security screening.

Table 3: Accelerated Algorithms and Data Structures results.

Measure	Result
Curated learning capsules	6
Initial public assessment	Passed
Initial sealed assessment	Correctly rejected
Failure attribution	Dijkstra precondition
Targeted remediation	Passed
Fresh sealed assessment	Passed
Generated property cases	120/120
Dependency-planning transfer	Passed
Restart reconstruction	Passed
Security screening	Passed
Unsafe actions/live writes	0/0

The resulting academy receipt is verified and the module is **passed_bounded**. Advanced Python is the active next module; Testing/Debugging is unlocked, whereas later systems and security modules remain prerequisite-locked. This is bounded algorithms apprenticeship, not a claim of comprehensive algorithms expertise.

1.6 Governed Intelligence Glyph Consolidation

Unbounded retention of copied documents, lessons, and trajectories would cause working memory to grow without limit. AION therefore separates immutable evidence from compressed active intelligence. A governed glyph is represented as

$$g = (i, k, n, d, E, s, r), \tag{8}$$

where i is a content-derived identifier, k the glyph kind, n the canonical name, d compact structured data, E a set of evidence identifiers, s the epistemic state, and r a Tessaris/GlyphOS-compatible rendering.

Three operational glyph classes are retained:

Knowledge verified concepts and bounded rules;

Skill verified proposal-only executable procedures; and

Curriculum proposed learning material explicitly marked `proposed_not_learned`.

The system stores content-addressed references rather than copying raw evidence into every glyph. Before expansion, the source artifact must still exist and its SHA-256 digest must match. A representative human rendering is

`<Knowledge|dijkstra_nonnegative_precondition:g_<content-id>->Verify>`.

The actual implementation uses the canonical Tessaris double-bracket Unicode delimiters in the stored rendering; the ASCII presentation above is used only for portable typesetting.

Let B_E be the bytes required to copy all admitted source artifacts into working memory and B_G the compressed glyph packet size. The working-memory reduction is

$$R_G = 1 - \frac{B_G}{B_E} = 0.970815 = 97.08\%. \quad (9)$$

Table 4: Governed intelligence-glyph consolidation results.

Measure	Result
Verified source artifacts	6
Knowledge glyphs	25
Skill glyphs	6
Curriculum glyphs	13
Total glyphs	44
Working-memory reduction	97.08%
Provenance resolution	100%
Raw evidence copied into glyph store	No
Original evidence deleted	0
Curriculum glyphs marked learned	0
Checksum-tampered packet	Rejected

The compressed packet is not a substitute for evidence. It is a compact index of verified abstractions and procedures. The immutable evidence ledger remains the authority, while glyphs accelerate retrieval, composition, and future neural consolidation.

1.7 Persistence, Verification, and Claim Boundary

The background mastery service now prioritizes the depth-first academy, keeps broad curricula parked, and rebuilds glyph consolidation only when verified source artifacts change. The focused academy, algorithms, glyph, teacher-cycle, and autonomous-supervisor stack passed 16/16 tests. The glyph tests include checksum alteration and confirm fail-closed behavior.

These promotions establish a coherent and continuously operating accelerated learning substrate. They do not establish instant expertise, completion of the thirteen-module academy, general subject-executor invention, multi-month retention, or AGI. Advanced Python, Testing/Debugging, Software Engineering, operating systems, networking, security, distributed systems, production operations, and the integrated capstone remain to be earned through executable outcomes. The correct claim is therefore:

AION now possesses an evidence-backed North Star, an honest mastery ledger, a governed teacher and examiner separation, a persistent curriculum controller, a depth-first accelerated engineering academy, one newly completed bounded algorithms apprenticeship, and provenance-preserving compressed intelligence.